

HUMAN COMPUTER INTERFACES

Writing 3

MATHEMATIC MODELS FOR ANIMATION (2D-3D)

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LEAST SQUARES REGRESSION

Least Squares Regression's method is for calculate the function behavior of Simple Linear Regression. It predicts the next movements from data set of (x,y) or more spaces variables in animation

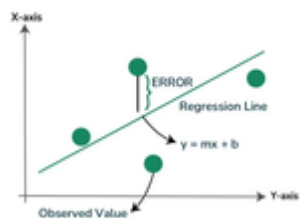
$$f(x) = mx + b$$

m = the range of change

x = next position

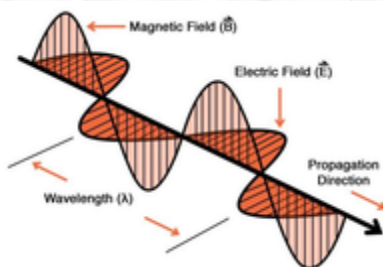
b = position where the object starts

Least Square Method



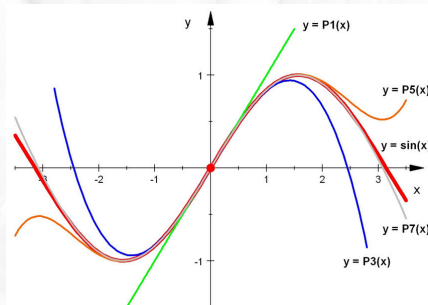
SINUSOIDAL MOVEMENT

sinusoidal movement modelates an oscillation from objects, this could used for trayectories or rebounds. It is a smooth change of position



TAYLOR SERIES

This model uses many elements to simulate a similar movement for a smooth movement and trayectories



$$\sum_{n=0}^{\infty} \frac{f^n(a)}{n!} (x-a)^n$$

CIRCLE'S EQUATION

For orbital motion, we need to make a circle where the body will move through, using sin and cousin parameters with the radius to change position respect time

$$x^2 + z^2 = r^2$$

